

**Time:** 5 minutes**Outcome:** Use real GDP per person and compounding to compare living standards.**Difficulty:** ●●○ Intermediate

Living Standards and Growth



THE CORE IDEA

Real GDP per person is a common measure of material living standards, and compounding makes sustained growth powerful over time. Dividing real output by population distinguishes growth in total production from growth in average material resources. Productivity growth can raise output per person without requiring proportionally more work, although this measure does not capture every dimension of well-being.



HOW TO RECOGNIZE IT

- A country raises output mainly by requiring much longer workweeks, while output per hour is unchanged.
- Higher productivity lets an economy produce more goods and services from its available labor and resources.
- Real GDP per person is commonly used to compare living standards and growth.
- Real GDP per person relates inflation-adjusted output to the number of people sharing that output.
- Compare growth in real GDP with population growth to distinguish total output from output per person.



WATCH OUT

A common mistake is to equate more total labor input with higher productivity or complete welfare improvement. Productivity is output per unit of input, and GDP omits important dimensions of well-being.



WORKED EXAMPLE: COMPOUNDING LIVING-STANDARD GROWTH

If real GDP per person grows at 4% per year, the Rule of 70 estimates a doubling time of $70/4 = 17.5$ years. This calculation describes material output growth, not complete well-being: longer work hours can raise GDP per person without raising output per hour or preserving leisure.



CHECK YOURSELF

At 4% annual growth, about how many years does the Rule of 70 predict for real GDP per person to double?

**READY?**

Return to the game and master this concept.